

Linear systems and elimination

Uniqueness of the right inverse minimizing an induced p -to-2 norm

Difficulty: hard **Importance:** interesting to specialist**Status:** Solved

Independently reviewed resolution - 2026-09-11

Negative resolution. Theorem 1 gives a 2×3 full-row-rank matrix with distinct norm-minimizing right inverses for every $2 < p < \infty$ over both fields. Their common induced norm is $2^{1/2-1/p}$. Sections 3–4 identify the complete minimizer sets, including a complex higher-dimensional family. The smallest matrix is attributed to Dokmanic and Gribonval's spectral-norm example; its extension to the displayed direct-inverse objective is checked. No conclusion about the separate product objective is claimed.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. [Complete manuscript](#), [independent proof review](#), and [submission record](#). The supplied notes were reconstructed with substantial AI assistance. This is independent agent verification, not external human peer review or formal proof-assistant certification; no novelty or priority claim is made.

The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Hard reflects a focused uniqueness question for a known norm minimizer; specialist impact is the characterization of generalized inverses under induced norms.

Problem statement

Let $1 \leq m < n$, let $A \in \mathbb{C}^{m \times n}$ have rank m , and let $2 < p < \infty$. For $X \in \mathbb{C}^{n \times m}$ define the induced norm

$$\|X\|_{p \rightarrow 2} = \sup_{y \in \mathbb{C}^m \setminus \{0\}} \frac{\|Xy\|_2}{(\sum_{i=1}^m |y_i|^p)^{1/p}}.$$

The Moore–Penrose inverse $A^\dagger = A^*(AA^*)^{-1}$ minimizes this norm among the right inverses X satisfying $AX = I_m$. Is it always the unique minimizer? Equivalently, must

$$AX = I_m, \quad X \neq A^\dagger \quad \implies \quad \|X\|_{p \rightarrow 2} > \|A^\dagger\|_{p \rightarrow 2}$$

hold for every such A, p, X ?

This is the direct-inverse uniqueness question in Dokmanić–Gribonval's Remark 4.1. The endpoints and the separate objective $\|XA\|_{p \rightarrow 2}$ are outside this statement.

Connection to numerical linear algebra

The norm measures the worst Euclidean solution amplification for right-hand sides bounded in ℓ^p . Uniqueness determines whether another linear solver can match the Moore–Penrose inverse's optimal amplification while changing its sparsity or structure.

References

1. I. Dokmanić and R. Gribonval, *Beyond Moore–Penrose Part I: Generalized Inverses that Minimize Matrix Norms*, arXiv:1706.08349v2 (2017), §2.1 and §2.4 (right inverses and A^+); §4.4, Corollary 4.2(3) and Remark 4.1, manuscript p. 18 (PDF page 18). [Preprint and version history](#). [PDF](#).
2. I. Dokmanić and R. Gribonval, *Part II: The Sparse Pseudoinverse*, 2017, §2. Its entrywise-norm objective is different.

Earlier status check — 2026-09-08

Checked the complete v2 statement and version history on 2026-09-08; v2, dated 2017-07-13, remains the latest arXiv version of Part I. Searches combined the authors, the paper title, “induced norm”, “p to 2”, “unique”, “uniqueness”, “generalized inverse”, and 2025/2026. The companion paper’s sparse-inverse uniqueness results concern entrywise norms and generic inputs, not this induced-norm claim for every full-row-rank matrix. No later proof or counterexample was located. This bounded search does not certify that no resolution exists elsewhere.

Audit update — 2026-09-10

The [author copy of Part I](#), Corollary 4.2(3) and Remark 4.1 on printed p. 18, explicitly separates minimality from the remaining uniqueness question for $2 < p < \infty$. Searches for later induced- p -to-2 uniqueness results found no resolution; Part II’s sparse-inverse objectives do not supply this missing assertion.